

Single Diaphragm Differential Pressure Gauge (SD)

The differential pressure gauge consists of high pressure and low pressure chambers separated by a diaphragm element. The pressure differential between the high pressure chamber and low pressure chamber leads to the axial deflection of the element transmitted to the movement through the linkage rod. The movement converts the axial deflection into the angular rotation of the pointer. It is designed for measuring differential pressure in tough conditions and is built with special consideration to high corrosive resistance. It has applications in various industries.

The use of corrosion resistant material such as high grade stainless steel, for both the casing and the measurement system, makes this pressure gauge best suitable to operate in an adverse environment and with corrosive media.

The welded design coupled with a modular construction system, allows for best in class ingress protection (upto IP-65 standard). The gauge has a long and stable service life with tried and tested safety features adhering to the EN 837-3 standard.



Available In

Nominal sizes	4"(100 mm), 6"(150 mm)
Ranges	250 mm WC to 10000 mm WC 0 to 0.6 bar upto 0 to 10 bar

Special Features

- Compliance to the latest EN837-3
- Overload capacity 110 % of FSD
- Ingress protection to IP - 65
- Accuracy class $\pm 1.6\%$ of FS
- Stainless steel casing
- Dry or liquid filled version
- Static pressure up to 40 bar double side

Industrial Applications

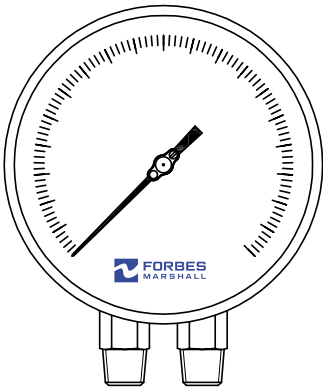
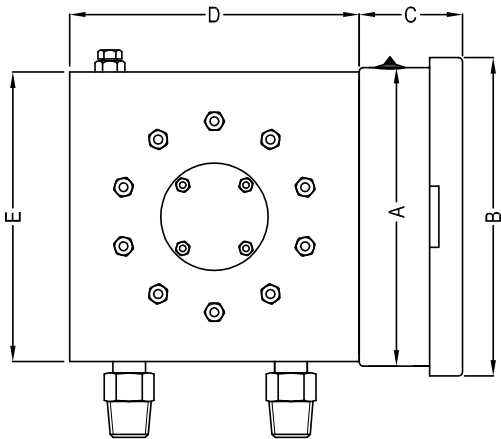
- Chemical and fertiliser industries
- Petrochemical industries
- Thermal, nuclear and hydroelectric power stations
- Pharmaceutical industries
- Food process industries
- Water treatment plants
- Rubber molding / processing plants
- Hydraulics and pneumatics
- Sugar industries
- Refineries
- Hydraulic power packs

Standard Product Specifications	
Specifications	Description
Reference standard	EN 837 - 3
Mounting	Direct bottom
Connection	1/4" NPT (F)
Case and bezel	SS 304 (bayonet type bezel)
Diaphragm	SS 316l
HP/IP chamber	SS 316
Movement	SS 304
Dial	Aluminum, white background with black marking
Pointer	Aluminum, black color micrometer with zero adjustable
Accuracy	±1.6% of FS (standard)
Over pressure limit	110% of FS (as per EN standard)
Window	Toughened glass
Protection	IP 65
Blow off disc and filling plug	Neoprene
Gasket	Silicon
Ambient temperature	-25°C to 65°C
Process temperature	Max. 200°C (for dry gauge)
	Max. 65°C (for glycerine filled gauges)

Standard Pressure Ranges				
Code	Kg/Sq.cm (Code A)	Bar (Code B)	Static Pressure	
AB	0 + 0.6	0 + 0.6	40 bar double side	
AC	0 + 1	0 + 1	40 bar double side	
AD	0 + 1.6	0 + 1.6	40 bar double side	
AE	0 + 2.5	0 + 2.5	40 bar double side	
AF	0 + 4	0 + 4	40 bar double side	
AG	0 + 6	0 + 6	40 bar double side	
AH	0 + 10	0 + 10	40 bar double side	
Code	MmWC (Code X)	Static Pressure	Code (Code - E)	Dual Scale Bar/PSI
AO	0 + 250	10 bar double side	BK	0 + 1/0 + 15
AP	0 + 400	10 bar double side	BL	0 + 1.6/0 + 23
AQ	0 + 600	10 bar double side	BO	0 + 6.0/0 + 86
AY	0 + 1000	10 bar double side	BP	0 + 10/0 + 150
AZ	0 + 1600	10 bar double side	BZ	0 + 2.1/0 + 30
BA	0 + 2500	10 bar double side	CA	0 + 4.2/0 + 60
BB	0 + 4000	10 bar double side	CB	0 + 7/0 + 100
BC	0 + 6000	10 to 40 bar double side		
BD	0 + 10000	10 to 40 bar double side		

Mounting Options

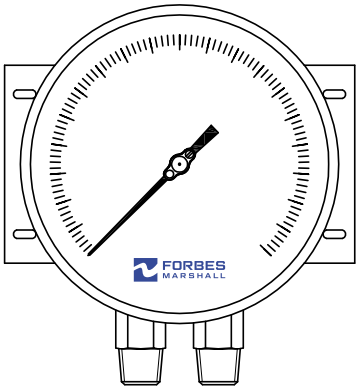
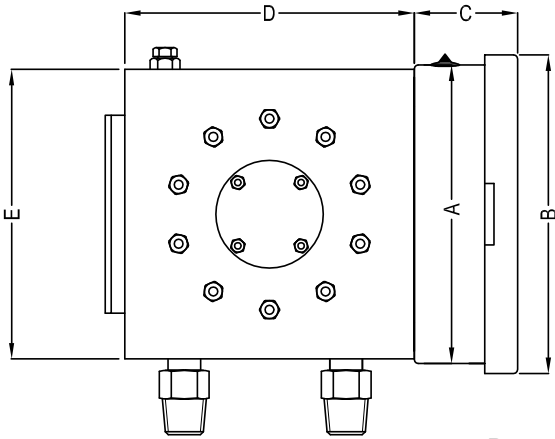
Direct Bottom



→ Parameters

Dial Size	A	B	C	D	E
100	100	110	48.5	105	100
150	150	160	50	150	150

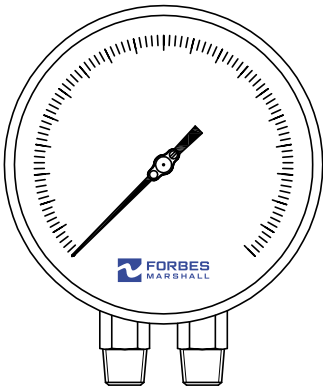
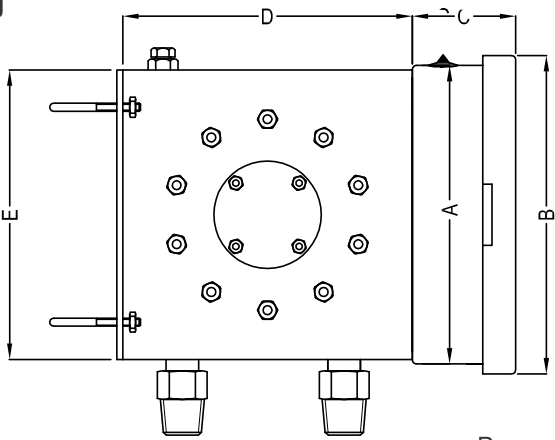
Surface Bottom



→ Parameters

Dial Size	A	B	C	D	E
100	100	110	48.5	105	105
150	150	160	50	150	150

2" - Pipe Mounting



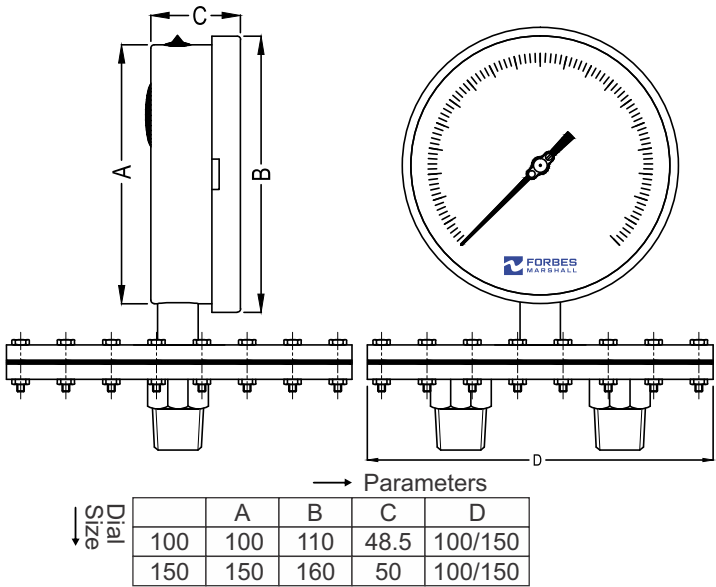
→ Parameters

Dial Size	A	B	C	D	E
100	100	110	48.5	105	105
150	150	160	50	150	150

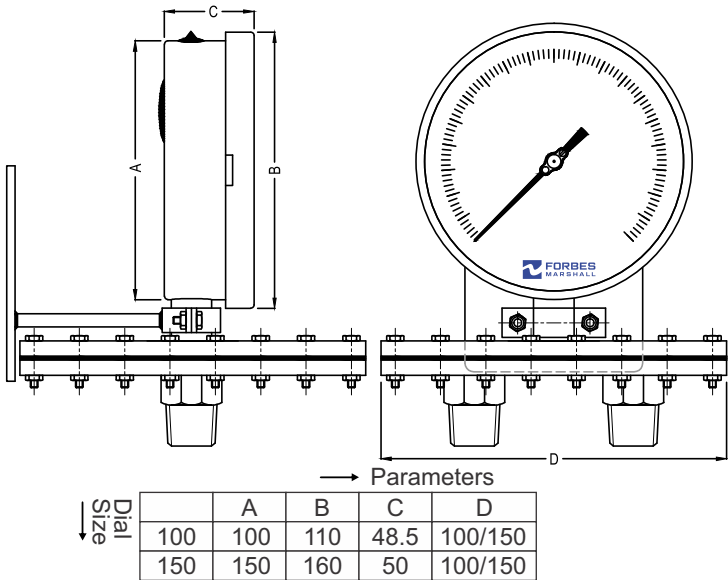
Note: Chamber size shall vary according to ranges

Mounting Options

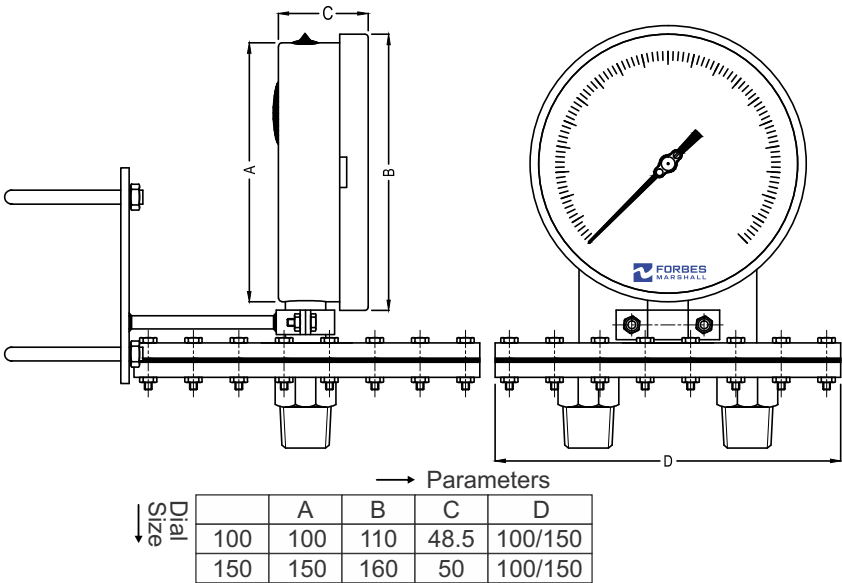
Direct Bottom



Surface Bottom



2" - Pipe Mounting



Note: Chamber size shall vary according to ranges

Ordering Information

Model		**	*	*	*	*	**	*
Single diaphragm differential pressure gauge	SD							
Dial Size								
100 mm	E							
150 mm	F							
Mounting								
Direct bottom	1							
Surface bottom	4							
Panel mounting bottom entry	A							
Panel mounting back entry	B							
Process Connection Size								
1/4"	A							
3/8"	B							
1/2"	C							
Process Connection Type								
NPT(M)	A							
BSP(M)	B							
BSPT(M)	C							
NPT(F)	D							
BSP(F)	E							
BSPT(F)	F							
Range								
Please refer standard pressure range table								
Engineering Units								
Kg/cm ²	A							
Bar	B							
Kpa	C							
mmWC	X							

Configuration Generation Code

Case and Bezel		*****			
SS 304 (standard)					
SS 316					
Movement					
SS 304					
SS 316					
HP/LP Chamber					
SS 316					
SS 316L					
Window					
Toughened glass					
Shatterproof safety glass					
Static Pressure					
Static pressure double side 5 bar					
Static pressure double side 10 bar					
Static pressure double side 15 bar					
Static pressure double side 20 bar					
Static pressure double side 25 bar					
Static pressure double side 40 bar					
Diaphragm - Material of Construction					
SS 316					
SS 316L					
Optional Features					
Case filling – glycerin upto 65°C					
(for ranges above 1 bar)					
Case filling – silicon Oil -40°C to 200°C					
(for ranges above 1 bar)					
Tag marking on dial					
2" pipe mounting bracket – SS 304					
2" pipe mounting bracket – SS 316					
Wall mounting plate					
External zero adjustment					
Custom Designed dial					
Special requirement – please specify					
Tests and Certificates					
Five point calibration certificate (standard)					
Material test certificate					
NACE traceability certificate					
CE marking as per 97/23/EC					
Ultrasonic testing					
IBR certification					
EN 10204 3.1.B certificate					
Helium leak test					
Oxygen service					
Special requirement – please specify					

Ordering example: SDE1ADAFB-XXXX (with standard selection)



Forbes Marshall
Krohne Marshall
Forbes Marshall Arca
Codel International
Forbes Solar
Forbes Vyncke
Forbes Marshall Steam Systems

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